

Chapter 1

Preliminaries

1.1 Basic Covering Properties

We use the usual terms and notations^[1]. An ordinal is the set of all smaller ordinals, and a cardinal is an initial ordinal. We use κ, μ for cardinals; λ a for infinite cardinal; $\alpha, \beta, \gamma, \theta$ for ordinal numbers. $\omega = \{0, 1, 2, \dots\}$ is the smallest infinite cardinal. $|A|$ denotes the cardinality of a set A . The empty set ϕ is the unique set B such that $|B| = 0$. $\mathbb{R}$ denotes the real line $[0, 1]$ the closed unit interval in $\mathbb{R}$.

For any sets Z, Y , we denote $\mathcal{P}(Z) = \{A : A \subset Z\}$; $Y^X = \{f : f \text{ is a function from } X \text{ into } Y\}$. If $f \in Y^X$, we write $f : X \rightarrow Y$.

If $f : X \rightarrow Y$, $A \subset X$ and $B \subset Y$, then $f(A) : A \rightarrow Y$ is defined by $f|_A(x) = f(x)$ for each $x \in A$; $f[A] = \{f(x) : x \in A\}$. $f^{-1}B = f^{-1}[B] = \{x \in X : f(x) \in B\}$.

For a set A and $n < \omega$, let $[A]^n = \{B \subset A : |B| = n\}$, $[A]^{<\omega} = \cup\{[A]^n : n < \omega\}$. Let $\omega^0 = \{\phi\}$. For $n = 1, \dots, \omega$, let $\omega^n = \{f : f; n \rightarrow \omega\} = \{(k_0, k_1, \dots, k_{n-1}) : k_i \in \omega, i < n\}$. If $h < \omega$, let $\phi \oplus h = (h)$, $(k_0, \dots, k_{n-1}) \oplus h = (k_0, \dots, k_{n-1}, h)$. Let $\omega^{<\omega} = \cup_{n < \omega} \omega^n$.

Let $\mathcal{U} = \{\mathcal{U}_\alpha \in \Delta\}$ be any family of subsets of X . Without loss of generality we may assume the indexing set Δ be any set of ordinals. We call